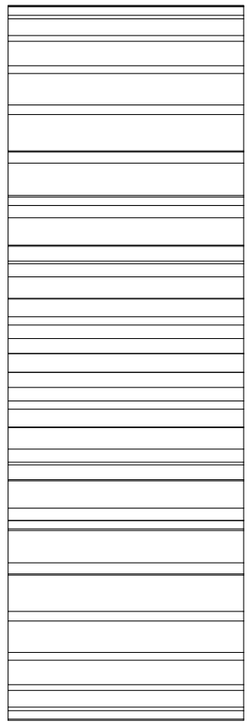
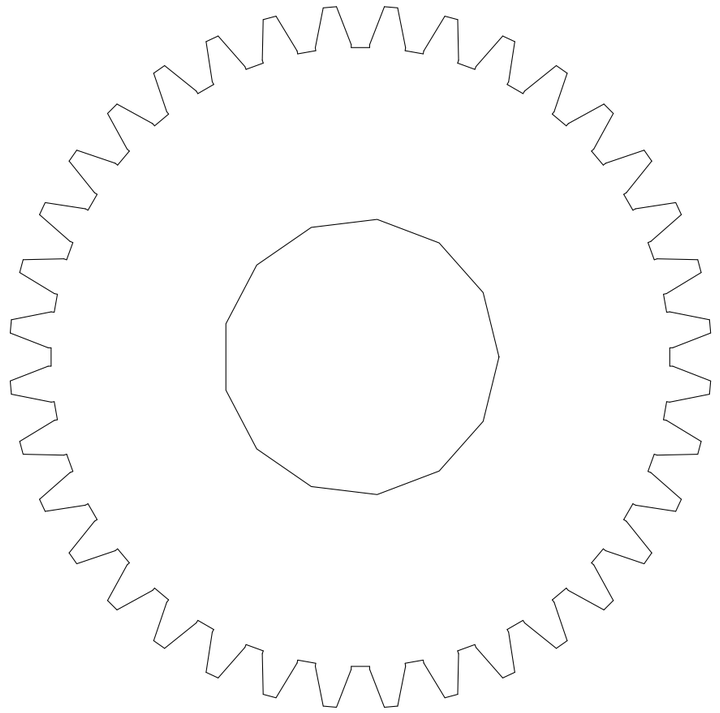
VAYN shoulder · milling drawing · 24 Planet-1

FRONT · Z horizontal, +X up



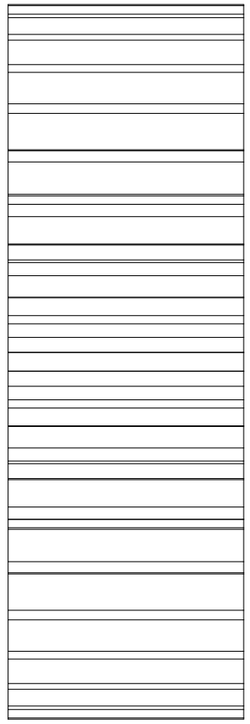
scale 3.84:1 view span 10.0 × 30.3 mm

END · from the arm side (+Z)



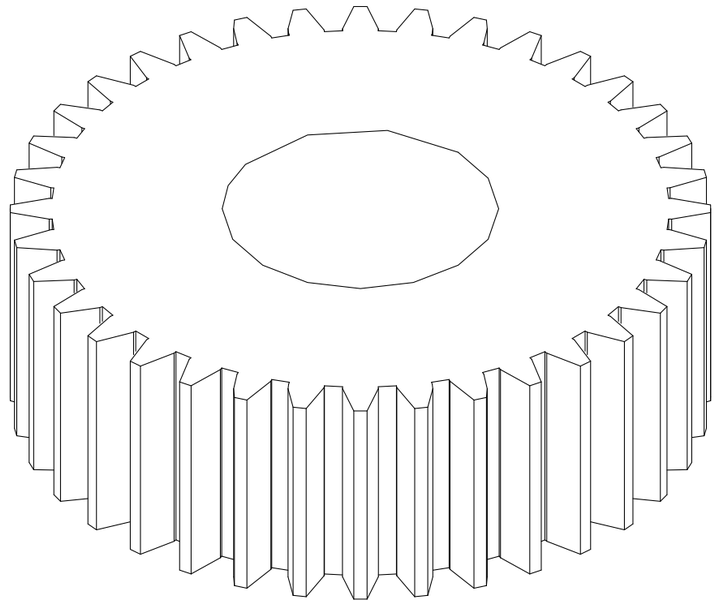
scale 3.77:1 view span 30.3 × 30.3 mm

SIDE · Z horizontal, +Y up



scale 3.84:1 view span 10.0 × 30.3 mm

ISOMETRIC



scale 3.76:1 view span 30.4 × 25.7 mm

24

Planet-1

Quantity in this actuator: 3
Material: 20MnCr5
Group: Gears
Finished size: Ø30.4 × 10 mm face, bore Ø12 mm
Mass: 41.61 g each
Model bounding box, X × Y × Z: 30.3 × 30.3 × 10.0 mm
Raw stock: Ø35 × 120 mm 20MnCr5 bar, same heat. Hob the three blanks as one stack.
Process: Turn, hob, part off
Heat treatment / finish: Carburize, case 0.12–0.18 mm, 58–62 HRC. Grind the bore after hardening.
STEP solid: 24.1 Planet-1 (20MnCr5)

Controlled features

- 36 teeth, module 0.8. Span over 5 teeth W5 = 11.008 mm, minus 0.015 / plus 0.
- ISO 1328 grade 7–8 is accepted. Do not grind the teeth.
- Bore Ø12 N6, roundness IT5/2, Ra 0.8 µm or finer.
- Three identical parts. In the STEP they are 24.1, 24.2 and 24.3.

STEP file is the geometric master. Do not scale this sheet. Sizes not listed here follow ISO 2768-1 class m and ISO 2768-2 class K. Drawing 24_planet_1.pdf